Artavazd Maranjyan

arto.maranjyan@gmail.com
artomaranjyan.github.io
Google Scholar Linkedin

Current Position

2026–present **Postdoctoral Researcher and EPFL AI Center & Swiss AI Initiative Fellow,**
EPFL, Lausanne, Switzerland
Jointly hosted by **Volkan Cevher**'s LIONS group and **Martin Jaggi**'s MLO group.
Developing theoretically grounded methods to make large-scale AI training faster and more resource-efficient, with emphasis on distributed and asynchronous optimization, communication efficiency, and model parallelism.

Education

2023–2025 **Ph.D. in Computer Science**, *KAUST*, Thuwal, Saudi Arabia; **Advisor:** Peter Richtárik
Thesis: First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data

2021–2023 **M.Sc. in Applied Statistics and Data Science**, *Yerevan State University*, Yerevan, Armenia
Thesis: On local training methods; **Co-supervisors:** Peter Richtárik, Mher Safaryan

2017–2021 **B.Sc. in Informatics and Applied Mathematics**, *Yerevan State University*, Yerevan, Armenia
Thesis: On the Convergence of Series in Classical Systems; **Supervisor:** Martin Grigoryan

Research Experience

03/2025 **Visiting Researcher, hosted by Yi-Shuai Niu**, *BIMSA*, Beijing, China
- Collaborated on server-assisted federated learning.
- Delivered invited seminars at Peking University (PKU), Beihang University (BUAA), and BIMSA.

04/23–08/23 **Researcher, Martin Grigoryan's group**, *Yerevan State University*, Yerevan, Armenia
- Investigated universal functions for the Vilenkin and Haar systems across different function spaces.

06/22–01/23 **Research Intern, Peter Richtárik's group**, *KAUST*, Thuwal, Saudi Arabia
- Contributed to GradSkip (TMLR), a communication-efficient local optimization method with improved complexity.

01/22–08/23 **Machine Learning Researcher**, *YerevaNN*, Yerevan, Armenia
- Researched optimization methods for federated learning.

Industry Experience

07/21–06/22 **Co-Founder**, *OnePick*, Yerevan, Armenia
- Built an MVP that helped content creators and marketers select photos matching their text descriptions.
- Advanced through FAST's InVent 2.0: team formation, ideation, product development, and pitching.
- Secured initial funding and continued development beyond the program.

07/21–09/21 **Backend Developer**, *EXALT Technologies Ltd*, Yerevan, Armenia
- Contributed to backend development for a Nutanix project.

06/21–07/21 **Machine Learning Research Engineer**, *FAST*, Yerevan, Armenia
- Developed machine-learning models for fraud detection.
- Built machine-learning and statistical models for data-driven forecasting.

09/19–01/21 **Software Engineer in Test**, *Picsart*, Yerevan, Armenia
- Developed test-automation solutions across several mobile and web applications.
- Collaborated across engineering, QA, product, and business teams to diagnose and resolve product issues.
- Proposed improvements to DevOps processes and Jenkins CI pipelines.

Awards

06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowship**, *EPFL*
Competitive fellowship supporting independent postdoctoral research (CHF 5,000 in annual travel funding)

05/2025 **CEMSE Dean's List Award**, *KAUST*
Awarded for excellent academic and research performance (USD 2,500)

09/2023 **Dean's Award**, *KAUST*
Merit-based award for selected incoming KAUST students (USD 18,000 over three years)

05/2021 **Outstanding Final Project Award**, *Yerevan State University*
Recognized for an outstanding bachelor's thesis (one of 6 recipients among 250+ students)

Publications

15. "LOSCAR-SGD: Local SGD with Communication-Computation Overlap and Delay-Corrected Sparse Model Averaging." Y. Maziane, A. Mahran, A. Maranjyan, P. Richtárik. *arXiv:2605.20866*, 2026.
14. "Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method." A. Sadiev, A. Maranjyan, I. Ilin, P. Richtárik. *arXiv:2605.18174*, 2026.
13. "Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity." A. Mahran, A. Maranjyan, P. Richtárik. *arXiv:2605.13434*, 2026; [poster].
12. "Rennala MVR: Improved Time Complexity for Parallel Stochastic Optimization via Momentum-Based Variance Reduction." Z. Tovmasyan, A. Maranjyan, P. Richtárik. *arXiv:2605.08871*, 2026.
11. "Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity." A. Maranjyan, P. Richtárik. ***ICLR** 2026*, *arXiv:2509.22860*; [poster] [slides].
10. "BiCoLoR: Communication-Efficient Optimization with Bidirectional Compression and Local Training." L. Condat, A. Maranjyan, P. Richtárik. *arXiv:2601.12400*, 2026.
9. "ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning." A. Maranjyan, E. M. Saad, P. Richtárik, F. Orabona. ***ICML** 2025*, *arXiv:2502.00775*; [poster].
8. "Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity." A. Maranjyan, A. Tyurin, P. Richtárik. ***ICML** 2025*, *arXiv:2501.16168*; [poster] [slides] [video].
7. "MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times." A. Maranjyan, O. S. Omar, P. Richtárik. ***UAI** 2025*; *OPT 2024*, Oral (top 5%), *arXiv:2410.04285*; [poster] [slides] [video].
6. "GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity." A. Maranjyan, M. Safaryan, P. Richtárik. *TMLR 2025*, *arXiv:2210.16402*; [poster] [slides] [video].
5. "LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression." L. Condat, A. Maranjyan, P. Richtárik. ***ICLR** 2025*, Spotlight (top 5.1%), *arXiv:2403.04348*; [poster] [video].
4. "Differentially Private Random Block Coordinate Descent." A. Maranjyan, A. Sadiev, P. Richtárik. *OPT 2024*, *arXiv:2412.17054*; [poster].
3. "Menshov-type theorem for divergence sets of sequences of localized operators." M. Grigoryan, A. Kamont, A. Maranjyan. *Journal of Contemporary Mathematical Analysis*, vol. 58, no. 2, pp. 81–92, 2023.
2. "On the divergence of Fourier series in the general Haar system." M. Grigoryan, A. Maranjyan. *Armenian Journal of Mathematics*, vol. 13, pp. 1–10, 2021.
1. "On the unconditional convergence of Faber-Schauder series in L^1." T. Grigoryan, A. Maranjyan. *Proceedings of the YSU A: Physical and Mathematical Sciences*, vol. 55, no. 1 (254), pp. 12–19, 2021.

Reviewing

TMLR (2024–present); NeurIPS and ICML (2026); ICLR (2025); SIMODS and JMLR (2024).

Selected Talks and Poster Presentations

08/2026	Poster, ***Swiss Optimization Symposium***, Ascona, Switzerland [poster]
04/2026	**Invited talk**, ***Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)***, Rio de Janeiro, Brazil [video] [slides]
07/2025	**Talk** and poster, ***35º Colóquio Brasileiro de Matemática***, *IMPA*, Rio de Janeiro, Brazil [slides] [poster]
04/2025	**Invited talk**, *Federated Learning One World Seminar (FLOW)*, Online [video] [slides]
03/2025	**Invited talk**, ***Academic Report of the School of Mathematical Sciences***, *Beihang University (BUAA)*, Beijing, China, hosted by Jiaxin Xie [slides]
03/2025	**Invited talk**, ***Optimization Seminar***, *BIMSA*, Beijing, China, hosted by Yi-Shuai Niu [slides]
03/2025	**Invited talk**, ***Seminar***, *Peking University*, Beijing, China, hosted by Kun Yuan [slides]
11/2024	**Invited talk**, *Machine Learning Research at Apple*, Online, hosted by Samy Bengio [slides]
03/2024	Poster, ***The Machine Learning Summer School in Okinawa 2024***, *OIST*, Okinawa, Japan [poster]
10/2023	**Invited talk**, ***Algorithms & Computationally Intensive Inference seminars***, *University of Warwick*, Coventry, England [slides]
12/2022	**Invited talk**, *Federated Learning One World Seminar (FLOW)*, Online [video]